WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 6: Project: Procedural City

Module 2 | 50 minutes | Project: procedures | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. procedural generation ___ A. Testing one small piece on its own.
b. design document ___ B. A plan of procedures and parameters.
c. unit test ___ C. Creating content with code and randomness.

2. Plan building (height)

Order the building procedure Number them 1–5.

a. ___ Move down by height
b. ___ Move right 40
c. ___ Move up by height
d. ___ Pen down at the bottom left
e. ___ Pen up, ready for the next building

3. My plan

Plan your project: Generate a city skyline from procedures: a building procedure with a height parameter called in a loop with random arguments, plus at least one detail procedure. Plan it in a design document and test each procedure alone. Sketch it and list what each part will do.

4. Checklist

Tick each one when your project does it.

a. ☐ Design document
b. ☐ Procedures with parameters
c. ☐ Random arguments
d. ☐ 8+ buildings
e. ☐ Detail procedure
f. ☐ Tested

5. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ My city is built from procedures with parameters.
☐ It uses random arguments.
☐ I tested my procedures on their own.